

Electric Vehicle as fuel Alternative Prediction

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Abstract—In this paper, we explore the question of whether EVs can be considered a viable replacement for traditional cars that rely on fossil fuels. Looking at EV performance and energy efficiency, machine learning models such as Random Forest, Gradient Boosting and Multi-Layer Perceptron (MLP) etc. we will be implementing these models in this paper. The customer expectations of EVs were analyzed using a dataset from Washington State. Precision and F1-score metrics were used to compare the models. We are also going to find out the best-performing algorithms from Machine Learning Models. This paper demonstrates that EVs have a good chance of becoming a popular eco-friendly substitute, which is further supported by robust forecasting models and effective energy control policies.

Index Terms—Electric Vehicle, EV, Fuel, Battery, Alternative fuel, Sustainable Energy, Logistic Regression, Gaussian Naive Bayes, Decision Tree, K-Nearest Neighbors, Random Forest, AdaBoost, Gradient Boosting, Support Vector Machine, Multi-Layer Perceptron

I. INTRODUCTION

Electric vehicles (EVs) are emerging as sustainable alternatives to their internal combustion engine (ICE) counterparts. This has created an urgency in reducing greenhouse gas emissions and dwindling reliance on fossil fuels. The paper focuses on the role of machine learning (ML) in making EV technologies smart with optimization-related functionality, including battery management, range prediction, and energy consumption. Emerging areas are providing integration aspects between renewable energy and EVs, discussing the hindrances such as charging infrastructure and real-world data limitations. The current methodologies will be reviewed, and future approaches suggested to provide insights into advancing EV adoption, enhancing energy efficiency, and delivering solutions toward better sustainable transport systems worldwide.

II. LITERATURE REVIEW

Boudmen et al. [1] presented a discussion about the EV possibilities as internal combustion engine (ICE) vehicles are substitute and possibilities of energy consumption, range extension and battery disposal. Neural networks, gradient-boosting decision trees, and regression models are focused on due to their applicability to improve, or optimize, the efficiency of electric vehicles (EV) driven. Five problems including battery degradation, charging infrastructure, and real-time optimizing are presented. The research also looks at energy control methods, the recalculation of brakes on electric vehicles, and solar panels in electric cars. However, there is

still a lack of robust research in expressing ethical issues and data privacy concerns in ML applications, as a result, leading to limitations in the broader adoption of EVs.

Jui et al. [2] stated the review on energy management strategies (EMS) of hybrid electric vehicles (HEVs) with particular focus on the use of ML techniques to improve energy efficiency and emission performance. It categorized EMS approaches into rule base, optimization and learning base and mentioned some of the ML algorithms like support vector machines, artificial neural networks, reinforcement learning for optimal power distribution. According to the findings, cases with the application of ML-based approaches can be regarded as more effective than those based on traditional approaches in terms of efficiency, the ability to adapt to real-time conditions, and above all, in extending the battery life. Nonetheless, there remains insufficient real data, high computational costs, and complex models. The present work advises incorporating mixed strategies and enhancing data acquisition for future developments, thereby illustrating the revolutionary prospects of ML in enhancing HEV energy control.

In the study by Shil et al. [3], the authors use the machine learning algorithms of linear regression, random forest, and gradient boosting regressor to predict future adoption of the electric vehicles in the USA. The variables include economic, environmental, and policy variables collected from other agencies, for instance, the U.S. Census Bureau, Environmental Protection Agency, and Department of Energy. They indicate gradient boosting and random forest have higher accuracy of forecasts compared to linear regression. EV uptake drivers are incentives, infrastructure, and concerns to the environment. However, The enable has limitations which include the following assumptions of constant energy price and inadequate regional analysis. The study points to the need for a clear direction in terms of policies to support and the subsidies and infrastructure development necessary to foster a faster transition to electric cars.

The goal of this research was to address the issues that Battery Electric Vehicles (BEV) faces [4]. Arguably the biggest issue for BEV is charging time. One way to address it is by decreasing their need for recharge. As a result a model was proposed which can help users predict driving range accurately such that users can use the prediction to prepare their trip.

The data for this paper was collected from vehicle mounted sensors. They used historic data. Lagrange Interpolation was used to fill data gaps. Root mean square error is used to determine interpolation effect. Multiple linear regression is used for correlation analysis. Pearson simple correlation coefficient is used to measure the strength of the correlation between driving range and variable. Partial correlation coefficient is used for multivariable analysis. The proposed algorithm is GBDT, as it performs better than multiple linear regression, with help of RMSE, MAE loss functions. A methodology of real time data collection from sensors and then generating new features from those data to build the tree is also mentioned. To avoid over-fitting issues feature selection has been done, where independent features with major roles were retained. In GBDT, pruning was introduced to avoid memorization.

The paper titled *"Can electric vehicles be an alternative to traditional fossil fuel cars with the help of renewable energy sources towards energy sustainability achievement?"* [5] deals with the integration of electric vehicles (EV) and renewable energy sources (RES) as a solution for energy sustainability. It accurately summarizes the U.S. Department of Energy presentation on the economic and environmental advantages of modeling and policy simulation of the electric vehicle adoption case, especially recalling EV case studies related to climate change and carbon emission reduction. The paper also addresses some challenges such as grid stress caused by high penetration of EVs and inefficient charging infrastructure. According to the paper, the PRISMA methodology was used to study the modern trends in electric vehicle technology, and possible ways to improve the efficiency of RES in biodiesel thermodynamic hybrid with electric vehicle (EV) charging stations, hydrogen systems, energy management systems optimization, and concordant EV charging were presented. The paper offers fitting conclusions on the use of EVs in promoting sustainability and a greener future for the automotive industry, although it also recognizes several issues regarding the optimal interaction between EVs and RES that require further research.

The paper *"Electric Vehicle Energy Demand Prediction Techniques: An In-Depth and Critical Systematic Review"* [6] represents a deep analysis by the authors on methodologies related to the prediction of energy demand associated with electric vehicles. The paper categorizes these approaches into three groups: linear, nonlinear, and hybrid models, providing a critical analysis of their strengths, weaknesses, and applicability. Key insights are drawn on challenges imposed by stochastic EV electricity consumption patterns and integrating those patterns within energy grids. The paper also identifies deficiencies in the existing literature, such as a lack of availability of real-world data and underreported sensitivity analyses. This work contributes significantly to the advancement of sustainable EV integration by providing a robust framework for future research and practical energy management strategies.

The study [7] focuses on estimating the driving range

of electric vehicles using machine learning. The study explores machine learning techniques for estimating driving range, addressing driver anxiety. A comprehensive dataset is constructed from various publicly available electric vehicle datasets. Utilizes different regression models, including linear regression and ensemble stacked generalization methods for accuracy. Regression techniques enhance driving range prediction accuracy, improving planning and reducing uncertainty. Evaluated against traditional heuristic approaches, machine learning models show promising predictive performance. Linear regression outperforms others in efficiency, accuracy, and practical implementation potential. Standard evaluation metrics like MAE, MSE, RMSE with cross validation were used to minimize bias. Machine learning techniques significantly improve electric vehicle driving range estimation compared to heuristic methods. Integration with real time datasets and further model refinement to enhance prediction capabilities is planned.

III. DATA PREPROCESSING & MODEL DESCRIPTION

Dataset is synthetically generated and focused on Washington State in the USA. Goal of the study is to help car manufacturers to judge if their electric car is a suitable fuel alternative vehicle to launch in Washington or not. Similar models can be scaled and tuned to a bigger audience.

Dataset has in total 17 features and 134474 instances. It is an imbalanced dataset with 3 different target classes. We had redundant features like State, DOL Vehicle ID and several location features to pinpoint the car registration location. Since the study is focused on Washington only, we can ignore the State completely, DOL Vehicle ID basically uniquely identifies each instance which has no impact on our classification model. Census Tract is a very inaccurate method of pinpointing location thus we removed it. County denotes a very large area so we removed it as well. After further analysis we found that 97.46% of our Base MSRP feature instances are corrupted. So we decided to drop it as well. The cars in the dataset featured two different types of battery technology, BEV and PHEV. They are inherently different and thus need different ways to measure their suitability. Thus we divided the Electric Range feature into two different features, combining Electric Range for BEV and Electric Range for PHEV. This ensured that the missing values in our Electric Range features were imputed using a more standardized value based on their specific Battery Type. The dataset had an uncleaned Electric Utility feature. We cleaned it to properly reflect the actual Electric Company names.

Since each of our Categorical features had a lot of nominal unique instances. We measured the correlation between each categorical feature pair using Cramér's V association rule. This provided us with the strength of association between categorical pairs. Afterwards we measured the Mutual Information of the pair values that had strong correlations against our target. The feature with the higher information was selected. We

found that the City name had strong correlation with the Postal Code, Legislative District and Electric Utility. Similarly Postal Code has strong correlation with City, Legislative District and Electric Utility. Legislative District has strong correlation with City and Postal Code. The Electric Utility has strong correlation with City and Postal Code. Car Make (Manufacturer name) had full correlation with the Model name. So among these City, Model and Postal Code was chosen. Model Year had no considerable correlation with any other categorical variables, so it was picked as well.

Postal codes consist of 5 digits and all States share the same first two digits. We dropped the first two digits of all postal addresses. Then divided the other 3 digits into three different features. We calculated Haversine distance for all instances using the provided Latitude and Longitude value. We considered a center point. By doing feature selection and feature engineering we managed to reduce dimension considerably during the encoding process. Since the target classes were imbalanced we took a stratified approach to split our data. To avoid data leakage all imputation encoding and scaling was done after splitting. We used median to impute the missing Electric range values as the outlier was significant. Separate median was considered for PHEV and BEV vehicle types. We used one hot encoding method on our categorical data, then scaled the values using Standard Scaler. After that we converted the matrix into a dense matrix, then applied principal component analysis to reduce the dimensionality of the matrix thus avoiding the curse of dimensionality. We transformed the data into 18 feature vector spaces with cumulative explained variance of 66%. Our data was ready to train. We trained the data on Logistic Regression, Gaussian Naive Bayes, Decision Tree, K-Nearest Neighbour, Random Forest, Adaptive Boosting, Gradient Boosting, Support Vector Machine and Multi-Layer Perceptrons. The performance metrics included accuracy, precision, recall, f1 score and their support along with a confusion matrix.

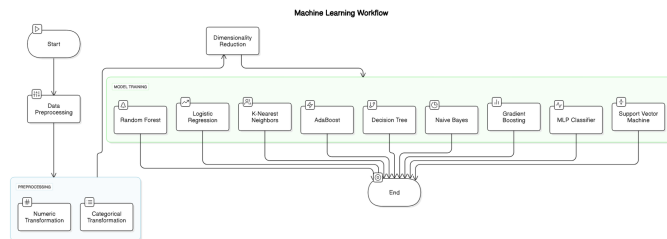


Fig. 1. Flow Chart of Fuel Alternative Prediction Using Machine Learning Models

A. Logistic Regression

Logistic regression is primarily used for binary classification. However for multi-class scenarios it uses multinomial logistic regression using a softmax function to model the likelihood of all classes simultaneously. The class with the highest probability is selected as the output class.

B. Gaussian Naive Bayes

Assuming conditional independence and normal distribution of feature vectors, Gaussian Naive Bayes uses Bayes Theorem to probabilistically classify. Used for its computational efficiency, performs best when strong independence among features is present.

C. Decision Tree

Decision tree splits a feature vector into subsets based on feature values. Starts with root features and goes to a leaf node while increasing the purity of each subsets. Widely used for its interpretability. Decision tree works great with multi-class problems, especially with imbalanced classes.

D. K-Nearest Neighbors

KNN works by mimicking the majority class of the nearest neighbors. The number of nearest neighbors for consideration depends on the k parameter while different distance metrics are used depending on the problem. Best used on low dimensional non overlapping datasets.

E. Random Forest

Based on Decision Tree, random forest is an ensemble learning algorithm. Works by a process called Bootstrapping which creates multiple sub-datasets from the main datasets then using multiple decision trees on those sub-datasets. Classify usually by majority voting. Usually performs better than Decision Tree and has reduced overfitting issues.

F. AdaBoost

Adaptive Boost works by using several weak classifiers to combine and create a strong classifier. For the Decision Tree it uses decision stumps. Each iteration a higher weight is assigned to misclassified instances and a lower weight is assigned to correctly classified instances. As a consequence the model is forced to focus on hard to classify instances.

G. Gradient Boosting

Uses a loss function to optimize weak learners. Residual error is used to minimize error compared to previous iterations. Each model is built after another with this correction until the loss function reaches minima or iteration level hits. Best for complex data.

H. Support Vector Machine

Uses support vectors to draw a hyperplane between data points. Hyperplane is drawn in such a way that maximizes the margin distance between data points.

I. Multi-Layer Perceptron

Feedforward artificial neural network, uses input, hidden and output layers to build the neural structure. Backpropagation is used to update bias and weight. Based on gradient descent to adjust weights and biases by reducing the loss function during backpropagation. Can be considered a black box algorithm for lack of explainability at bigger scale.

IV. RESULTS & ANALYSIS

Accuracy, macro Precision and Macro F1- Score is presented of all 9 listed machine learning models in the table. Since the dataset is imbalanced, macro F1-Score will be the prioritized measurement metric.

TABLE I
COMPARISON OF MODEL PERFORMANCE

Name	Accuracy	Macro Precision	Macro F1
Logistic Regression	95%	96%	94%
Gaussian Naive Bayes	83%	78%	78%
Decision Tree	96.4%	97%	95%
KNN	97.1%	98%	96%
Random Forest	98.2%	99%	97%
AdaBoost	95%	91%	92%
Gradient Boosting	96.8%	98%	95%
SVM	97%	98%	95%
MLP	98.4%	99%	98%

Among our models, MLP and Random Forest overall performed the best. Gaussian naive Bayes performed the least, proving our assumption that the features are strongly independent or normally distributed incorret. MLP, SVM and Gradient Boosting works great on complex non linear data. Since PCA was used for feature reduction and cosine metric ensured high accuracy for KNN. AdaBoost and Logistic Regression had unsatisfactory results compared to other models because Logistic Regression inherently works best with binary classification and AdaBoost suffered from overlapping data issues. Random forest robustly handled these issues with overall second highest accuracy.

V. LIMITATIONS & FUTURE RESEARCH SCOPE

The best way to improve the model is by incorporating data from other states, if possible from other countries as well. Our Base MSRP feature had 97% and Electric Range had about 50% missing values. Ensuring that the data are properly available can be a significant improvement. Delving into complex deep neural networks can also increase the accuracy at the cost of interpretability and complexity. Future research can also include vehicles with newer battery technologies. Overall this study can be expanded and tuned for specific companies to increase profit margin by deploying cars that fill certain criteria to benefit from legislative rules and benchmarks.

VI. CONCLUSION

The analysis came to the conclusion that policy-makers and electrical engineers can develop electric vehicles that outperform traditional fuel-dependent vehicles in terms of utility in social scenarios. Learning was integrated into the engineering process of EVs to address issues with vehicle efficiency and energy consumption. The effectiveness of MLP and Random Forest models for complex and nonlinear datasets was highlighted by the tests I ran. The analysis was further enhanced by tackling problems such as data quality and dataset unbalance. The results suggest that sustainable EV manufacturing strategies can be devised using advanced modeling and data-driven approaches. More work is required to extend the

analytics to larger data sets and fine tune these forecasts based on real-time data integrating.

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